Sinking Fund Method

Example 1:

Given: $C_0 = 56{,}000 + 4000 = 60{,}000$, $L = 16$ yrs, $C_L = 10\%$ or $60{,}000 = 6{,}000$
$i = 12\%$ or 0.12

Required: Book value at end of 12yrs (C_{12}) by SLM and SFM

Solutions:

SLM: $d = \dfrac{C_0 - C_L}{L}$

$d = \dfrac{60{,}000 - 6{,}000}{16} = 3{,}375$

$C_{12} = C_0 - 12d$

$C_{12} = 60{,}000 - 12(3{,}375) = \boxed{C_{12} = 19{,}500}$ SLM

SFM: $D_n = (C_0 - C_L)\left[\dfrac{(1+i)^n - 1}{(1+i)^L - 1}\right]$

$D_{12} = (60{,}000 - 6{,}000)\left[\dfrac{(1+0.12)^{12} - 1}{(1+0.12)^{16} - 1}\right]$

$D_{12} = \underset{A}{30{,}481.61881}$

$C_{12} = C_0 - D_{12}$

$C_{12} = 60{,}000 - A = \boxed{C_{12} = 29{,}518.38119}$ SFM

Example 2

Given: $C_0 = 150{,}000$, $C_L = 10\%$ or $150{,}000 = 15{,}000$, $L = 7$ yrs, $i = 10\%$ or 0.10

Required: Depreciation table for year 1 and 2

Solution: $d = (C_0 - C_L)\left[\dfrac{i}{(1+i)^L - 1}\right]$

$d = (150{,}000 - 15{,}000)\left[\dfrac{0.10}{(1+0.10)^7 - 1}\right]$

$d = 14{,}229.74246$

Year 2 Depreciation:

$14{,}229.74246 \times 0.10 = 1{,}422.974246$

Yr 2 d: $14{,}229.74246 + 1{,}422.974246$
$= 15{,}652.71671$

$d_n = d(1+i)^{n-1}$

$14{,}229.74246(1+0.10)^1 = 15{,}652.71671$

Year	Initial Book Value	Depreciation	total Depreciation	Book Value
1	150,000	14,229.74246	14,229.74246	135,770.2575
2	135,770.2575	15,652.71671	29,882.4596	120,117.5408
3				

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